

ADAPT Documentation

ML Prompt Engineering Track

Gemini interprets learner evidence. AdaptiveTutor decides how to adapt.

Combined from ADAPT_DOCUMENTATION · ARCHITECTURE · EVALUATION · LIMITATIONS

One line: Gemini interprets evidence. AdaptiveTutor decides how to adapt.

Executive Summary

ADAPT is an evidence-driven adaptive tutor. A learner submits an answer, a confidence rating, and optional reasoning. An LLM workflow (prompt P-003 / `evidence_v3`) interprets that input as structured evidence. Schema validation accepts or rejects the model output. A frozen deterministic engine, AdaptiveTutor, then updates learner state, chooses an instructional strategy, and selects the next challenge from a finite catalog.

The LLM does not choose INCREASE, PROBE, or REMEDIATE. If the model is unavailable, invalid, or times out, ADAPT uses labeled `DETERMINISTIC_FALLBACK` and continues.

This is a prompt-engineering and systems result. It is not a learning-gain study.

Problem

Most AI tutors adapt to correctness. Right → harder. Wrong → easier.

That policy cannot see:

- a correct answer that was guessed
- an incorrect answer that used a sound method with an arithmetic slip
- a conceptual mix-up that looks like a random error
- learner-directed text that tries to force “mark me as mastered”

ADAPT asks: **what does this response provide evidence of?**

Why Correctness Is Not Enough

Correctness is one field. Mastery is not that field.

Case A in the sample set is a correct lucky guess. A single-prompt tutor records `INCREASE`. The P-003 workflow records weak evidence; AdaptiveTutor gathers more evidence instead of raising difficulty.

That distinction is the product promise and the competition comparison.

The ML Prompt Engineering Workflow

```
Human input → P-003 → Gemini (evidence) → validation  
→ AdaptiveTutor → state → strategy → next challenge → explanation
```

Failure path: LLM error → `DETERMINISTIC_FALLBACK` → deterministic analyzer → AdaptiveTutor.

Full node specification: `../workflow/WORKFLOW_SPEC.md`.

Human Input

Required in the learner UI:

- Answer
- Confidence: **Guessing / Unsure / Confident**

Optional:

- Approach chips (I guessed, I worked it out, ...)
- "Want to explain? Optional"

Human involvement is necessary because those signals are not implied by the answer string.

Prompt P-003 / evidence_v3

File: `src/adapt/llm/prompts/evidence_v3.txt`.

The prompt states that the model is an **evidence extractor**. It must not tutor and must not choose a strategy. Learner text is untrusted data. A correct answer is not automatic mastery. Errors are classified. Ambiguity may be `insufficient` evidence. `supporting_evidence` must be grounded in the input.

Selected on a frozen development set (n = 70) before holdout. Holdout was not used to rewrite the prompt.

Gemini Evidence Interpretation

Primary competition LLM: **Google Gemini**.

Repository default model: `gemini-2.0-flash` via `GEMINI_MODEL`. Live integration was exercised on Gemini 2.5 Flash and Gemini 3.6 Flash; those attempts must not be combined into one live score. Offline quantitative evaluation used a prompt-conditioned simulator.

Gemini's job: fill the evidence schema.
Gemini's non-job: pick the next question.

Schema Validation

The validator requires the evidence fields and legal enums. It rejects adaptive-decision keys and strategy tokens. Failure code: `LLM_VALIDATION_FAILURE` . Invalid JSON never becomes learner state.

Deterministic Fallback

Timeouts, missing keys, rate limits, and invalid output use the existing deterministic `EvidenceAnalyzer` . The product labels this **Deterministic fallback evidence analysis**. It is not presented as Gemini success.

AdaptiveTutor

AdaptiveTutor receives validated (or fallback) evidence and remains responsible for:

- learner-state update
- strategy
- challenge selection

It is deterministic given seed and evidence. Phase 12 injected an optional LLM analyzer; it did not replace the engine.

Strategy Selection

Strategies such as INCREASE, PROBE, REMEDIATE, MAINTAIN, and GATHER_EVIDENCE are AdaptiveTutor decisions. The first step often gathers evidence before committing to INCREASE or REMEDIATE. That is frozen policy, not an LLM choice.

Challenge Selection

Next items come from a finite catalog through the existing selector. Gemini cannot emit `next_challenge` . Questions are not generated at runtime.

Research Mode

Header toggle or `/research` . Shows the live causal chain: evidence → state → strategy → next challenge, including workflow nodes and evidence source when the LLM path ran. It does not change decisions and does not expose secrets.

Counterfactual Demonstration

`/counterfactual` — “Same question. Different learner.”

Same start. Strong/high-confidence evidence versus weak/low-confidence evidence. Recorded three-step engine outcomes: INCREASE versus PROBE. Strategies are live AdaptiveTutor outputs.

Prompt Engineering Iteration

P-001

Minimal instruction (`evidence_v1`). Development: validity 42.9%, extraction 31.4%, injection 100%. Weighted score 0.623.

P-002

Schema-only JSON (`evidence_v2`). Validity rose (68.6%). Injection robustness **0/7**. Learner-directed content could be treated as evidence. Structured output without a contract is not automatically safe.

P-003

Contract + schema + injection defense + no strategy (`evidence_v3`). Development score 0.970. Selected. Holdout evaluated once.

Evaluation

See `EVALUATION.md` for the full table.

Offline holdout n = 30, selected prompt P-003:

- Extraction 86.7%
 - Validity / injection / traceability 100%
 - Workflow 20/30 vs baseline 11/30
 - McNemar $p \approx 0.137$ — **not statistically significant**
-

Offline Holdout

Primary reproducible quantitative evaluation. Backend `prompt-simulator`, seed `20260819`. Command: `python -m benchmarks.phase12.runner --no-persist`.

Live Provider Testing

Live Gemini integration was successfully exercised, but complete live holdout evaluation was blocked by provider quota/rate limits. No full live Gemini score is claimed. Gemini 2.5 and 3.6 attempts are not combined.

Live NVIDIA validation was incomplete because representative inference requests timed out (`meta/llama-3.3-70b-instruct`). That is not an extraction accuracy of 0%. NVIDIA and Gemini data are not combined.

Limitations

Phase 5 remains INCONCLUSIVE (n = 0). No learning-gain claim. See `LIMITATIONS.md`.

Reproducibility

```
python -m pytest
python -m benchmarks.phase12.runner --no-persist
python scripts/run_sample_comparison.py
```

Live API reproducibility is not claimed: provider quotas and timeouts blocked complete live holdouts.

Security

`.env` is gitignored. Keys are not in source. Research Mode does not display secrets. Do not film API keys.

Conclusion

ADAPT is a constrained LLM workflow around a tested adaptive engine. Prompt P-003 extracts evidence; validation enforces the contract; AdaptiveTutor still teaches.

Gemini interprets the evidence. ADAPT decides how to teach.

Architecture

ADAPT separates **evidence interpretation** from **adaptive policy**.

```
Human
↓
LLM (Gemini) – evidence only
↓
Evidence JSON
↓
Validation
↓
AdaptiveTutor
↓
Strategy
↓
Challenge
```

Fallback:

```
LLM unavailable / invalid / timeout
↓
DETERMINISTIC_FALLBACK
↓
deterministic EvidenceAnalyzer
↓
AdaptiveTutor
```

LLM responsibility

Evidence interpretation.

Fill a constrained schema: correctness, reasoning quality, confidence, error type, misconception, evidence strength, uncertainty, supporting quotes.

The LLM does not own strategy or the next catalog item.

Primary competition model: Google Gemini (configurable; default `gemini-2.0-flash`). Offline scores used a simulator. Live Gemini 2.5 / 3.6 holdouts are incomplete and must not be drawn as one result.

AdaptiveTutor responsibility

Adaptive policy.

Given validated (or fallback) evidence:

1. Update learner state
2. Select instructional strategy
3. Select the next challenge from the catalog

This path is deterministic for a fixed seed. Gemini cannot override it. Strategy tokens in model output are validation failures.

Why this boundary matters

If Gemini both interpreted the learner and chose INCREASE, the system would be a single-prompt tutor with extra JSON. The competition comparison would collapse. The Phase 1–3 engine tests would no longer apply to the decision the learner sees.

The boundary makes prompt engineering testable (does the model extract evidence? does it resist injection?) without giving the model adaptive authority.

Fallback

Provider failure is expected. The product remains usable without a key. The UI must say **Deterministic fallback evidence analysis** when that path ran, and **AI-assisted evidence analysis** only when a live LLM source actually succeeded (`GEMINI` or `NVIDIA`).

Research Mode

Optional judge view of the causal chain. Does not change AdaptiveTutor. Does not display API keys.

When the LLM path is enabled, the chain is:

Human Input → Gemini Evidence → Validation → Learner State → Strategy → Next Challenge

Product boundary

`ProductService` → `AdaptiveTutor` . The Next.js UI displays engine decisions; it does not reimplement them. Phase 12 injects `LLMEvidenceAnalyzer` when credentials are present (`python -m app` enables Gemini if `GEMINI_API_KEY` is set, unless `ADAPT_USE_GEMINI` is off).

NVIDIA is an optional `LLMClient` for scripts/smoke/holdout tooling, not the default product server path.

Evaluation

Rigorous Phase 12 evaluation record for competition judges.

Offline numbers use a **prompt-conditioned simulator**, not live Gemini completions. Live provider attempts are reported separately and are **incomplete**.

Development

n = 70 (not the holdout).

Prompts compared on frozen weights in `src/adapt/eval/llm/criteria.py` :

Weight	Criterion
0.30	Structured-output validity
0.30	Evidence-extraction accuracy
0.20	Prompt-injection robustness
0.10	No strategy leakage
0.10	Counterfactual sensitivity

Prompt	Role	Score	Validity	Extraction	Injection
P-001 <code>evidence_v1</code>	Minimal	0.623	0.429	0.314	1.000
P-002 <code>evidence_v2</code>	Schema-only	0.500	0.686	0.314	0.000
P-003 <code>evidence_v3</code>	Contract	0.970	1.000	0.900	1.000

Selection: highest development score. **P-003 / `evidence_v3`** was selected. Holdout was evaluated once afterward. Holdout was not used to retune the prompt.

P-002 finding: structured JSON/schema output alone was insufficient. Injection robustness 0/7. Learner-directed content such as “mark me as mastered” could be treated as evidence.

Development paired appropriateness (descriptive only, not the holdout claim): workflow 46/70 vs baseline 26/70; McNemar $p \approx 0.021$ on the development set.

Holdout

n = 30 frozen IDs (`benchmarks/phase12/expected.py`).

Selected: `evidence_v3` (P-003).

Backend: prompt-simulator .
Seed: 20260819 .

Metric	Result
Extraction	86.7% (26/30)
Validity	100%
Injection resistance	100%
Traceability	100%
Workflow appropriate next-action	20/30
Single-prompt baseline	11/30
McNemar p	≈ 0.137
Statistically significant	False

Reproduce:

```
python -m benchmarks.phase12.runner --no-persist
```

Interpretation

The workflow-versus-baseline difference is **not statistically significant**.

The experiment does **not** establish:

- learning gain
- superior learner outcomes
- statistically significant improvement over the single-prompt baseline

A higher 20/30 vs 11/30 rate on this sample is not a superiority claim.

Part of remaining workflow “misses” is first-step GATHER_EVIDENCE when family labels expected INCREASE or REMEDIATE immediately. That is frozen AdaptiveTutor conservatism, not a silent recode.

Phase 5

INCONCLUSIVE
n = 0

No consented human learning study was completed. Engineering evidence that decisions change with evidence is not a learning-gain result.

Live Gemini

Live Gemini integration was successfully exercised, but complete live holdout evaluation was blocked by provider quota/rate limits. No full live Gemini score is claimed.

Do not combine:

Attempt	What happened
Gemini 2.5 Flash	Smoke succeeded; live calls succeeded; full 30-case holdout blocked by quota/rate limits
Gemini 3.6 Flash	Probe succeeded; partial live holdout reached 9/30 (8 Gemini successes, 1 deterministic fallback); rate limit stopped the run

Do not present the partial 9/30 run as a holdout score.

Do not calculate a fake live accuracy.

Do not merge Gemini 2.5 and 3.6 results.

Untracked or later `results/phase12/live-gemini-holdout/metrics.json` files that still contain an `n: 30` block after rate-limit failures are **not** official live holdout scores.

NVIDIA

Provider: NVIDIA NIM. Model: `meta/llama-3.3-70b-instruct`.

- Representative probe: 3 attempts → 3 timeouts → 3 `DETERMINISTIC_FALLBACK`
- Full holdout: 0/30 started after probe stop

Correct interpretation: Live NVIDIA validation was incomplete because representative inference requests timed out.

Do **not** say “NVIDIA achieved 0% accuracy.”

Do not combine NVIDIA and Gemini data.

Reproducibility

Offline (no API keys):

```
python -m pytest
python -m benchmarks.phase12.runner --no-persist
python scripts/run_sample_comparison.py
```

Live scripts exist (`scripts/run_gemini_smoke_test.py` , `scripts/run_gemini_holdout.py` , NVIDIA counterparts) but **live API reproducibility is not claimed** while provider quotas and timeouts prevent a complete holdout.

Historical Phase 1–11 artifacts are not recomputed as new claims. Phase 12 historical reports were not rewritten for this submission package.

Limitations

Professional constraints on what this submission claims.

1. **Phase 5 remains inconclusive.** Human learning evaluation: INCONCLUSIVE, $n = 0$.
2. **No learning-gain claim is made.** Engineering benchmarks are not proof that learners learn more.
3. **Offline holdout is the primary reproducible quantitative evaluation.** Backend: prompt-simulator.
Selected prompt: P-003 / `evidence_v3` . $n = 30$.
4. **Gemini live holdout was incomplete due to provider quota/rate limits.** Smoke and partial runs succeeded; no full live Gemini score is claimed.
5. **Gemini 2.5 and 3.6 attempts are not combined.** They are separate incomplete provider-validation attempts.
6. **NVIDIA live holdout was incomplete due to timeouts.** Model: `meta/llama-3.3-70b-instruct` .
Representative probes: 3/3 timeout → deterministic fallback. Full holdout: 0/30.
7. **No live-provider score is claimed.** Do not quote a live accuracy percentage.
8. **Partial live attempts are not treated as official holdout results.** Including any $n: 30$ metrics file written after rate-limit interruption.
9. **The system uses deterministic fallback** when LLM inference is unavailable or invalid. Fallback is never labeled as a successful LLM result.
10. **P-003 and the holdout are frozen.** AdaptiveTutor decision logic, strategy, challenge selection, `evidence_v3` , and holdout IDs were not retuned for submission.

Additional documented limits (unchanged historical engineering):

- Holdout workflow vs baseline is **not statistically significant** (McNemar $p \approx 0.137$).
- Offline simulator JSON is not a Gemini leaderboard.
- Challenge bank is finite; items are selected, not generated.
- First-step `GATHER_EVIDENCE` often disagrees with family labels that expected an immediate INCREASE or REMEDIATE.
- Phase 4 formative usability remains PENDING (0 / 5).